To Reason but Not to Stop: Understanding and Recovering Pruned Large Reasoning Models

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Abstract

Weight pruning can degrade the accuracy of large reasoning models (LRMs), yet it remains unclear whether it damages reasoning itself. We find that a substantial fraction of this degradation stems from a failure to *stop* rather than to *reason*. As sparsity increases, pruned LRMs increasingly fail to terminate within the token budget, often after the correct answer has appeared in their reasoning, instead falling into repetitive loops, even though they still recognize where the dense model would stop. We further find that the hidden-state difference between terminating and looping continuations is consistent across pruned checkpoints, suggesting a shared direction associated with committing to an answer. Building on this, we propose Termination-Guided Activation Steering (TGAS), a training-free method that adds this direction to the residual stream at inference time. Across six LRMs, three pruning pipelines, and math and code benchmarks, TGAS improves termination and overall accuracy while reducing repetitive reasoning and generation length, showing that this failure is actionable.

1 Introduction

Large reasoning models (LRMs) achieve strong performance by generating long chain-of-thought traces, but this makes inference expensive (Zhang et al., 2026a; Guo et al., 2025). Weight pruning offers a natural approach to reduce cost by zeroing out selected parameters (Frantar and Alistarh, 2023; Sun et al., 2024). However, as consistently observed across pruning methods, pruning LRMs can substantially degrade accuracy on reasoning tasks (Ji et al., 2025; Lucas et al., 2026; Zhang et al., 2026b). Existing efforts have primarily focused on mitigating this degradation through improved pruning calibration (Ji et al., 2025; Lucas et al., 2026; Zhang et al., 2026b), alongside post-pruning recovery via retraining or fine-tuning (Liu et al., 2025b; Muralidharan et al., 2024; Kim et al., 2024; Xia et al., 2024a). Yet, it remains unclear whether this degradation reflects a loss of reasoning capability or a different kind of failure.

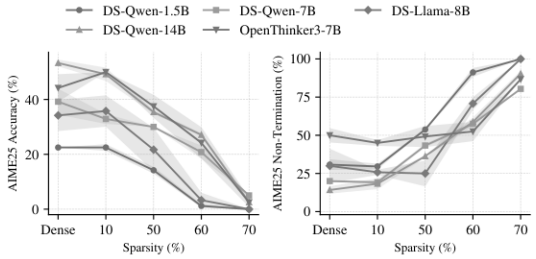


Figure 1: Accuracy (left) and non-termination rate (right) on AIME-25 across pruning settings (SparseGPT with OBC calibration, 16,384-token budget, averaged over 4 runs). Accuracy decreases as non-termination becomes more frequent.

We find that much of this degradation stems from a failure to *stop* rather than a failure to *reason*. As sparsity increases, pruned LRMs increasingly fail to terminate and commit to a final answer within the token budget, and accuracy falls accordingly (Figure 1). Strikingly, many of these non-terminating traces already contain the correct answer, yet the model keeps reasoning until the budget is exhausted. We refer to this behavior as ***reasoning divergence***. On problems solved by the dense model, pruned models remain largely accurate whenever they terminate, and non-termination becomes the dominant source of errors as sparsity grows (Section 3.1).

Why do pruned models fail to stop? Pruning does not erase the stopping signal: when replaying dense traces, pruned models still assign high probability to the end-of-thinking token where the dense model stops. Instead, during free-running generation they increasingly revisit earlier reasoning rather than moving forward, and fall into repetitive loops. Moreover, the hidden-state difference between continuations that terminate and

those that loop points in a similar direction across pruned checkpoints, suggesting that the shift between looping and committing to an answer is captured by a shared direction in representation space.

Based on this diagnosis, we propose *Termination-Guided Activation Steering* (TGAS), which adds this direction to the residual stream at inference time to steer pruned models toward termination without retraining or weight updates. TGAS consistently improves termination and accuracy across six LRMs, three pruning pipelines, and math and code benchmarks by reducing repetitive reasoning. Our contributions are threefold: (i) We show that a substantial part of pruning-induced degradation in LRMs arises from reasoning divergence, a failure to commit rather than a loss of reasoning ability; (ii) We identify a hidden-state direction that separates terminating from looping continuations, and build on it TGAS, a training-free steering method; (iii) We show that TGAS consistently improves termination and overall accuracy across models, pruning methods, and benchmarks by reducing looping.

2 Background and Related Work

Pruning for LRMs. One-shot unstructured pruning formulates compression as layer-wise reconstruction (Frantar et al., 2022): given a weight matrix $\mathbf{W}_\ell$ and calibration inputs $\mathbf{X}_\ell$, it solves $\min_{\|\hat{\mathbf{W}}_\ell\|_0 \le S} \|(\mathbf{W}_\ell - \hat{\mathbf{W}}_\ell)\mathbf{X}_\ell\|_F^2$, where S is the number of retained weights. We focus on two retraining-free methods: SparseGPT (Frantar and Alistarh, 2023), which uses second-order information to compensate for pruning errors, and Wanda (Sun et al., 2024), which scores weights by magnitude and input-activation norm. To better capture long reasoning traces, recent LRM pruning methods improve calibration via on-policy chain-of-thought RAC (Lucas et al., 2026), reasoning-step importance OBC (Nguyen et al., 2026), or self-generated traces (Ji et al., 2025; Wang et al., 2025; Xiang et al., 2025), detailed preliminaries in Appendix B.

Related Works. LRMs can overthink and enter repetitive loops (Chen et al., 2026; Pipis et al., 2026), which quantization can further prolong (Lotfi et al., 2026). We show that pruning amplifies looping while largely preserving the stopping signal, distinguishing failures to stop from failures to reason. Activation steering offers a way to modify behavior by adding directions to the residual stream, with difference-in-means directions shown to control behaviors of LLMs (Arditi et al., 2024). For reasoning, SEAL (Chen et al., 2025) suppresses reflection, while Plan-CoT (Sheng et al., 2025) modulates reasoning length. In contrast, TGAS targets pruning-induced non-termination using contrasting continuations of the same prefix, yielding a direction weakly aligned with SEAL's (Appendix F.8).

3 Understanding Pruning-Induced Reasoning Degradation

We investigate three questions: (i) Do pruned models lose reasoning ability or only the ability to stop? (ii) Is non-termination caused by limited generation budgets? (iii) Why do models fail to stop despite reaching the correct answer? We address the first two questions in Section 3.1 and the third in Section 3.2. Recall that we refer to this phenomenon as *reasoning divergence*, characterized by *non-termination*, driven by repetitive *looping* and an *failure to commit* to final answers.

Setting. We evaluate DeepSeek-R1-Distill-Qwen (DS-Qwen), DeepSeek-R1-Distill-Llama (DS-Llama) (Guo et al., 2025), and OpenThinker3 (OT3) (Guha et al., 2026) under one-shot unstructured pruning of decoder linear layers at varying sparsity levels, using SparseGPT or Wanda with OBC, RAC, or C4 calibration. Benchmarks include MATH-500 Levels 4-5 (hereafter MATH-500) (Lightman et al., 2024), AIME-25 (Zhang and Math-AI, 2025), and LiveCodeBench-v6 (LCB) (Jain et al., 2025); OpenR1-Math-220K (OpenR1-Math) (Hugging Face, 2025) is used only for calibration and steering-direction construction. We use a 16,384-token budget, $T = 0.6$, and top-$p = 0.95$, averaging $k = 4$ responses on three benchmarks. Our default uses SparseGPT with OBC calibration on model-generated reasoning traces, counting a response as correct only if it commits to a final answer, detailed in Appendix C.

3.1 Do Pruned Models Lose the Ability to Reason?

Accuracy on Dense-Correct Problems. To determine whether reasoning divergence reflects a loss of reasoning capability, we evaluate pruned models on problems correctly solved by their